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Fleming**

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(54) **SMART WRIST BAND**

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28, 2022.

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G06F 3/01 (2006.01)
G06F 3/16 (2006.01)
G06Q 20/40 (2012.01)
G06Q 30/0226 (2023.01)
G06Q 30/0251 (2023.01)
G08B 21/04 (2006.01)
G08B 25/12 (2006.01)

(52) **U.S. Cl.**

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(2013.01); **G06F 3/011** (2013.01); **G06F 3/167**
(2013.01); **G06Q 20/4016** (2013.01); **G06Q**
30/0226 (2013.01); **G06Q 30/0255** (2013.01);
G08B 21/043 (2013.01); **G08B 21/0446**
(2013.01); **G08B 25/12** (2013.01)

(58) **Field of Classification Search**

CPC G08B 25/016
See application file for complete search history.

(56)

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Primary Examiner — Travis R Hunnings

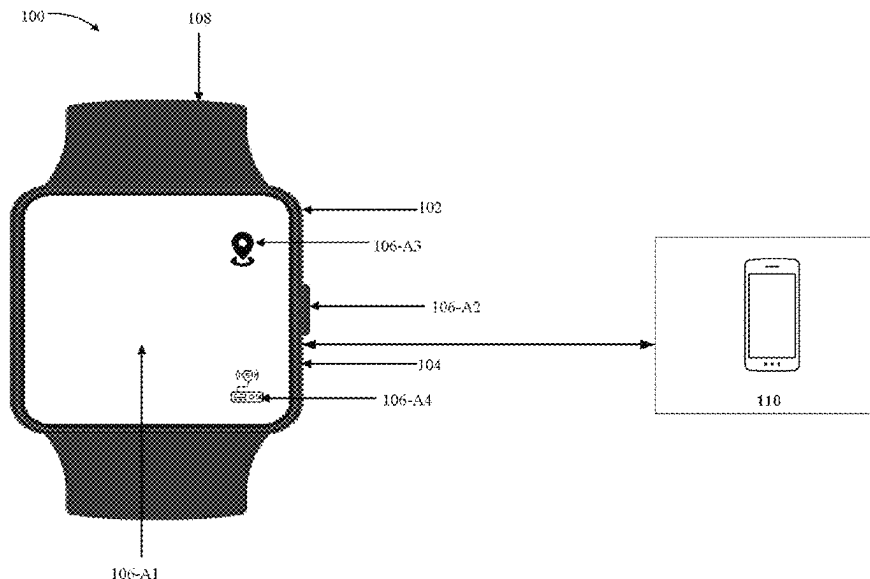
(74) *Attorney, Agent, or Firm* — Michael E. Zall

(57)

ABSTRACT

A user assistance system comprising a wearable device that enhances personal safety and health management. The wearable device includes an enclosure with a display screen to display a multidimensional code and an input interface for SOS signals. Additionally, the enclosure includes a geolocation unit and a data transceiver for communication with a mobile terminal. The mobile terminal scans the multidimensional code, establishes a communication link with the wearable device, and receives a user profile. The user profile includes emergency contacts, login details, and medical data. In emergencies, the system activates an SOS protocol, transmits a notification with the location of user to designated recipients, and facilitates response and assistance.

16 Claims, 12 Drawing Sheets



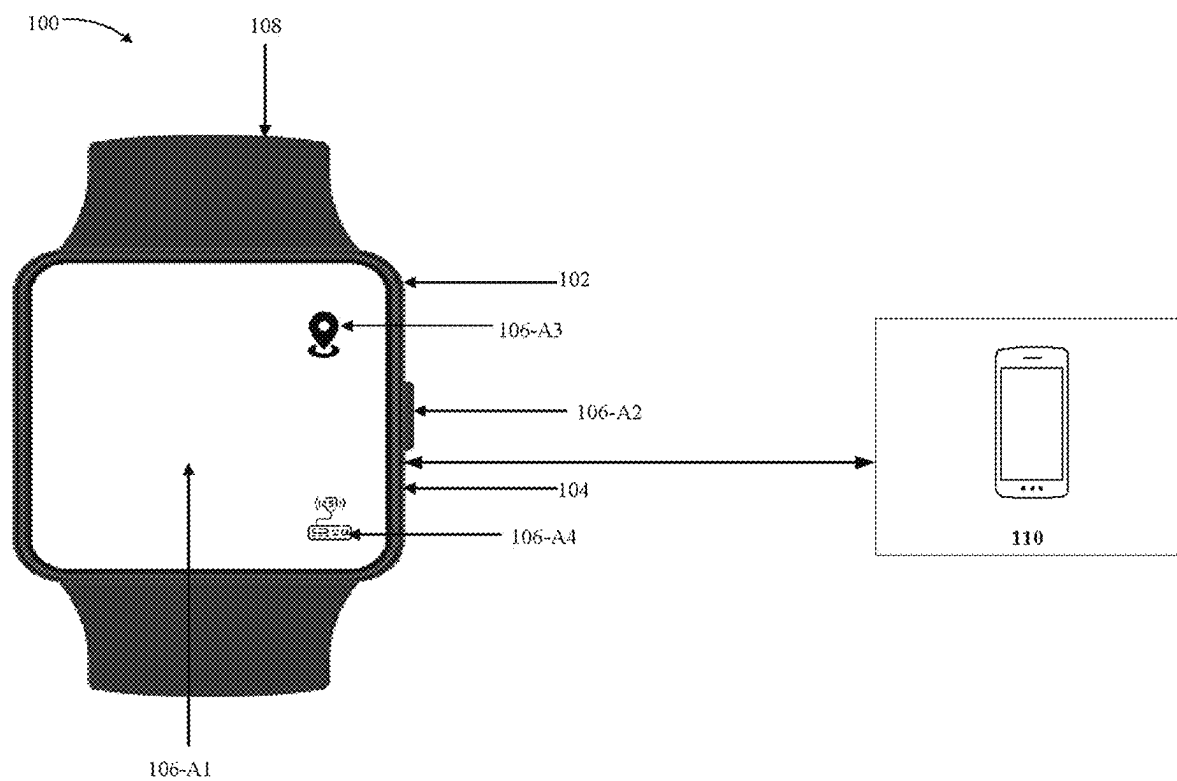


FIG. 1

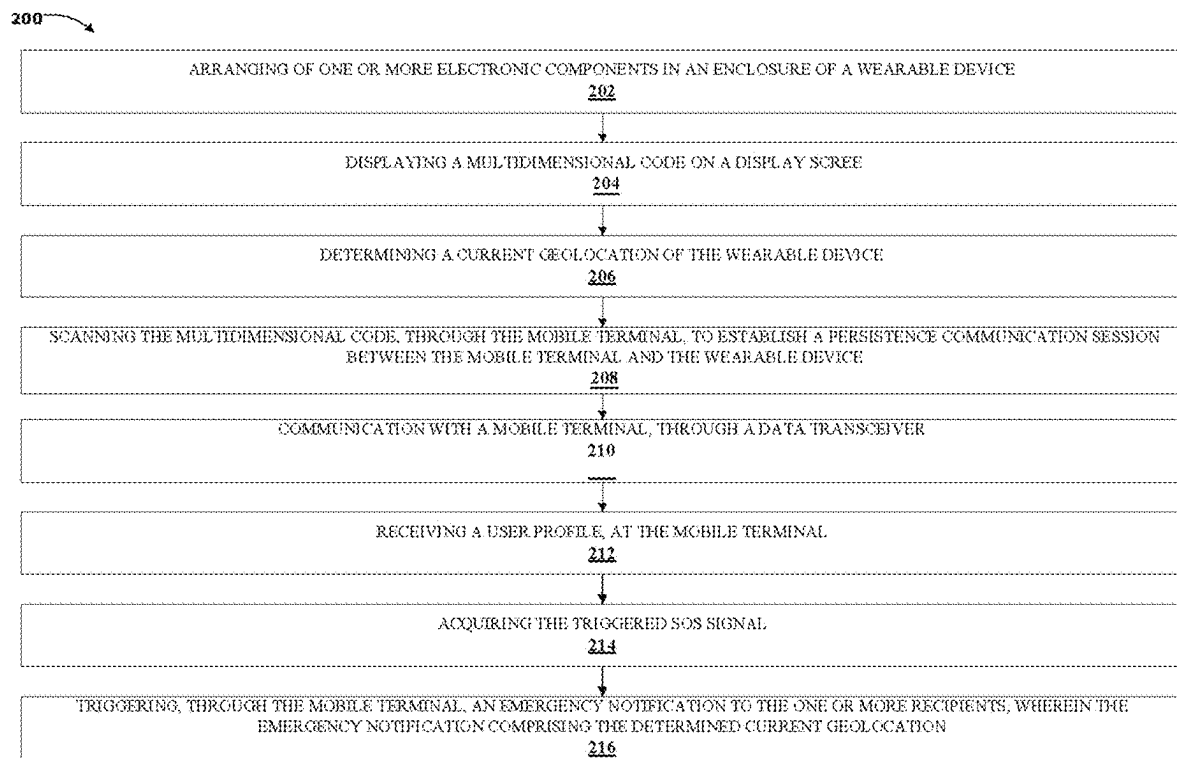
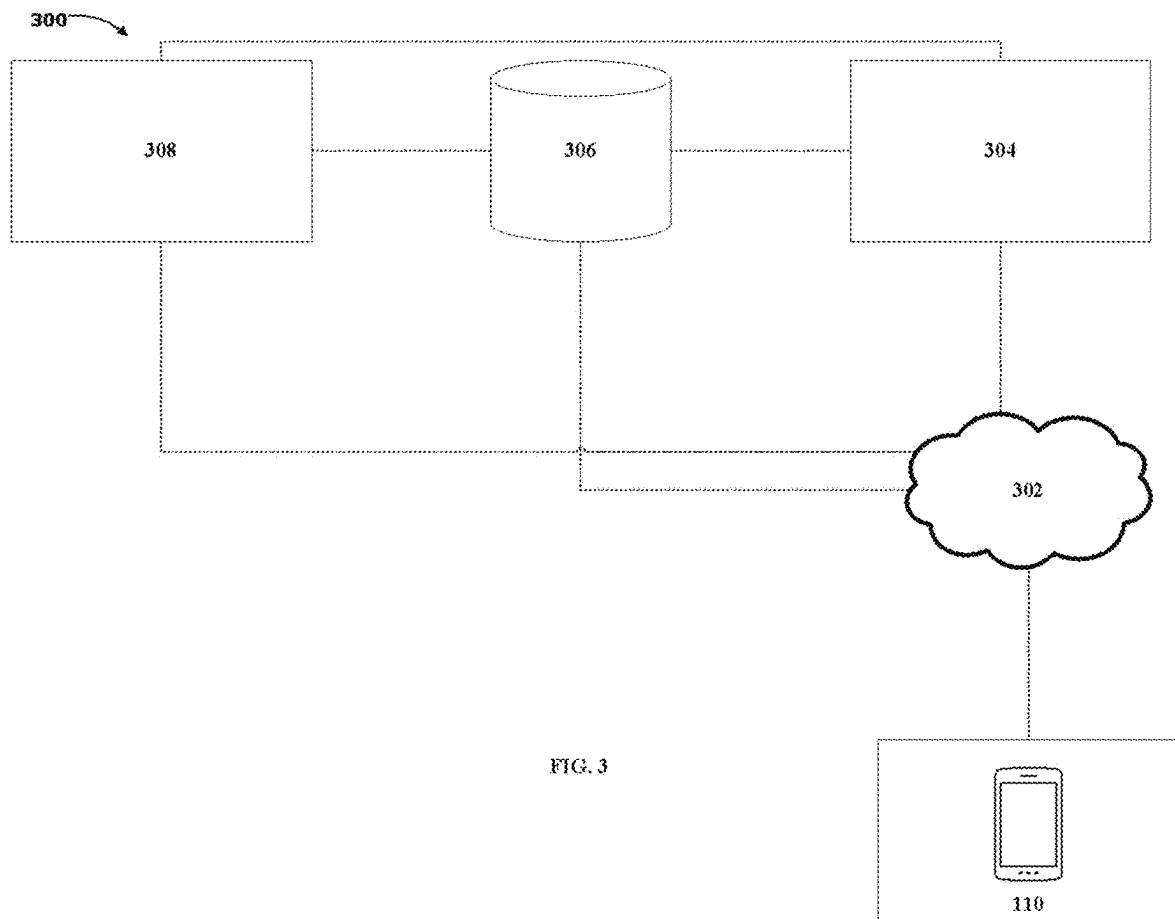


FIG. 2



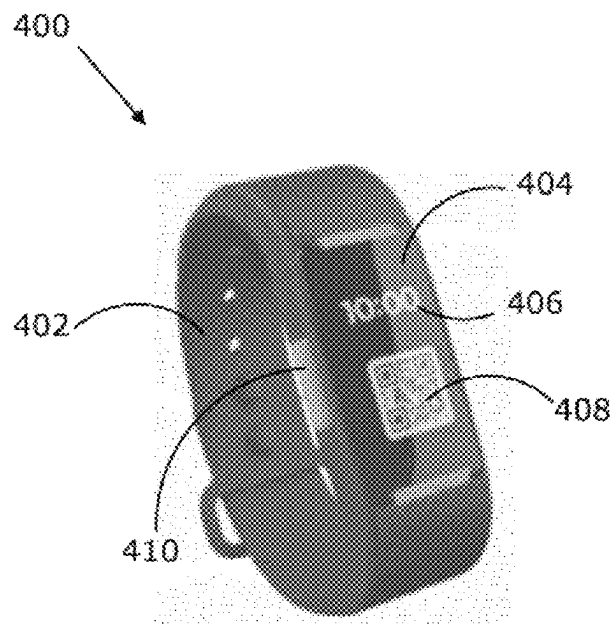


FIG. 4

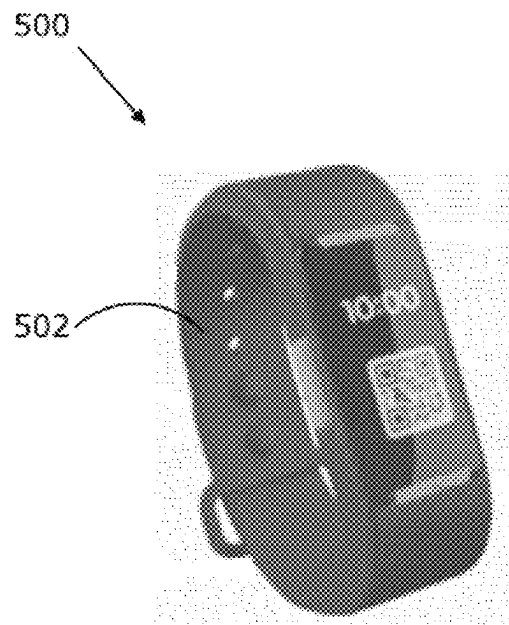


FIG. 5



FIG. 6

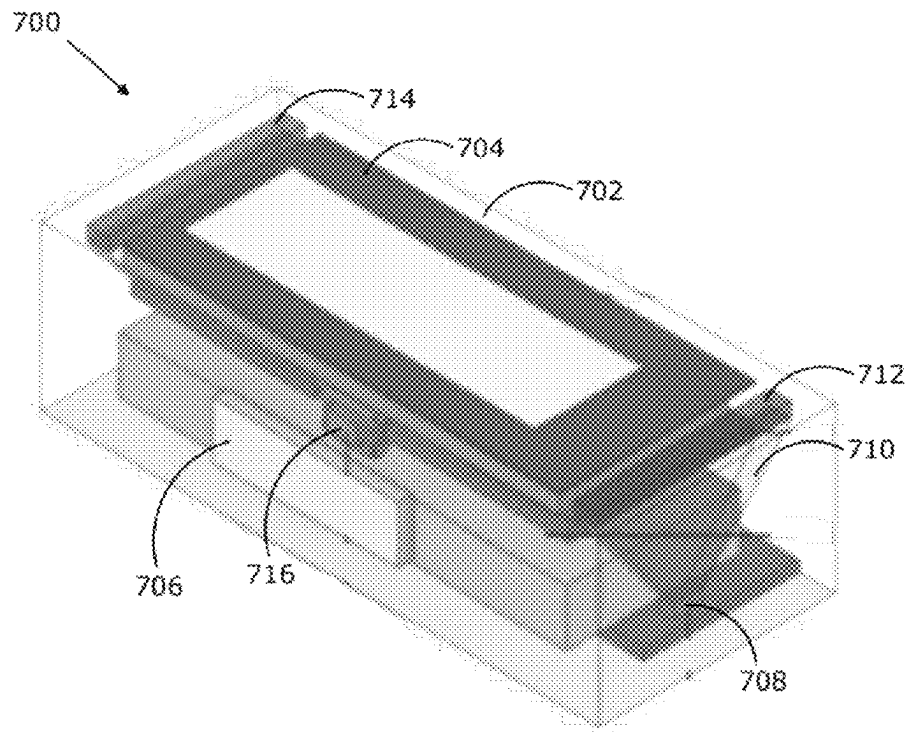


FIG. 7A

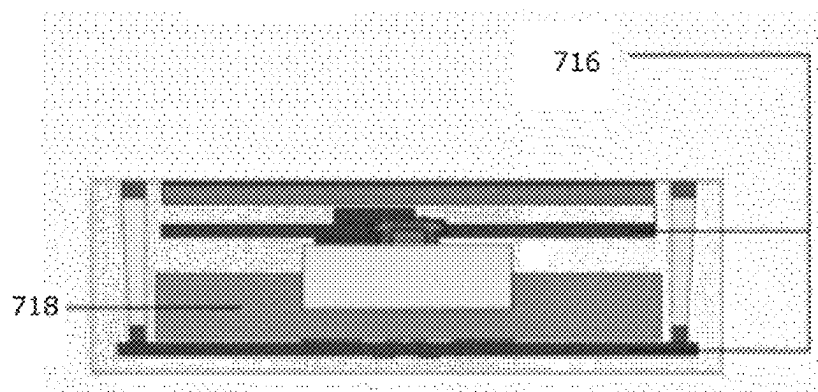


FIG. 7B

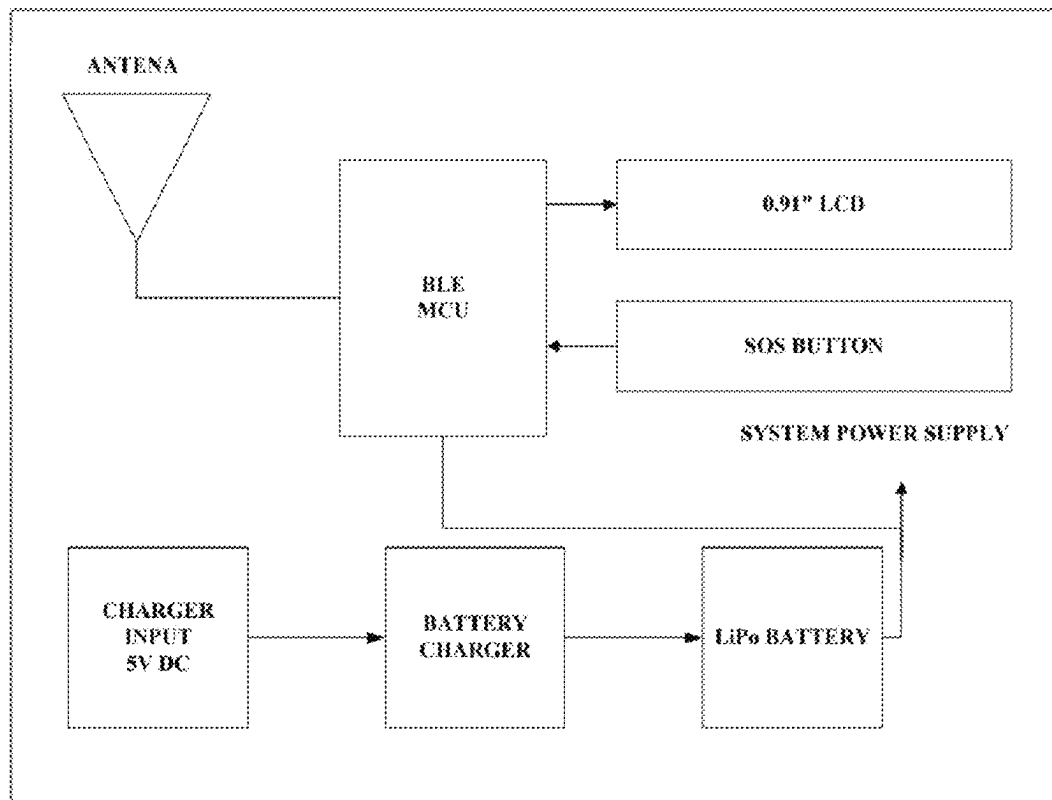


FIG. 8

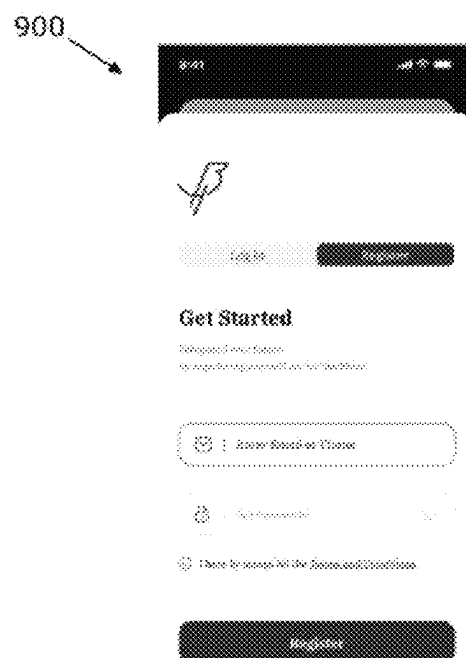


FIG. 9A

902

FIG. 9B

904

FIG. 9C

906

Black Bird Protection Community

Name (First/Last/Initials)

Address

City

State

Zip

Email

FIG. 9D

908

Black Bird Protection Profile

Name

John Smith

Phone

123-456-7890

About Me

I am a professional in the field of...

Contact Me

123-456-7890

FIG. 9E

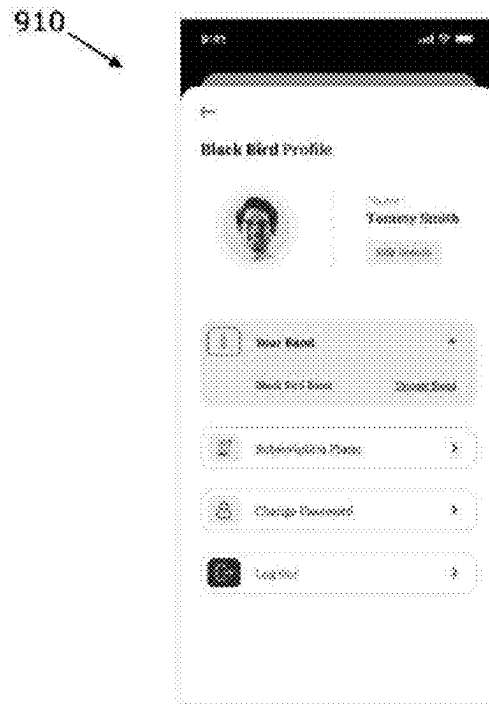


FIG. 9F

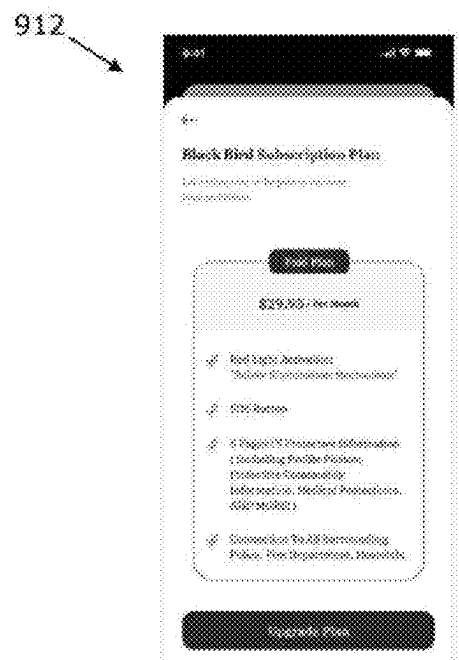


FIG. 9G

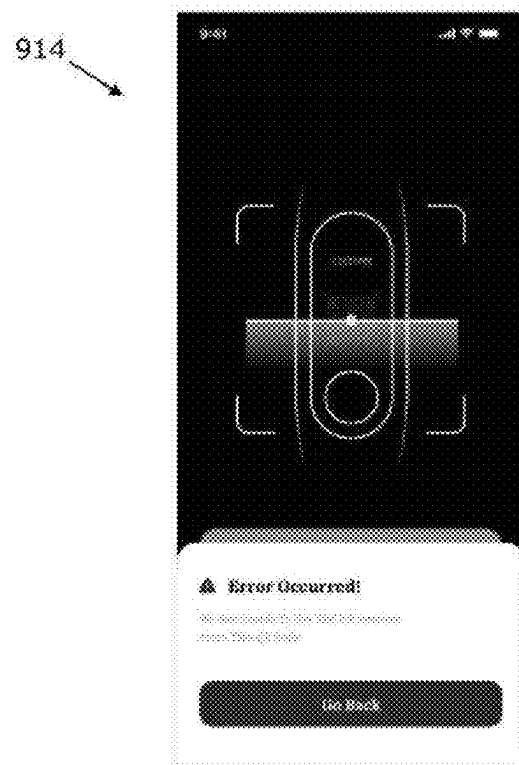


FIG. 9H

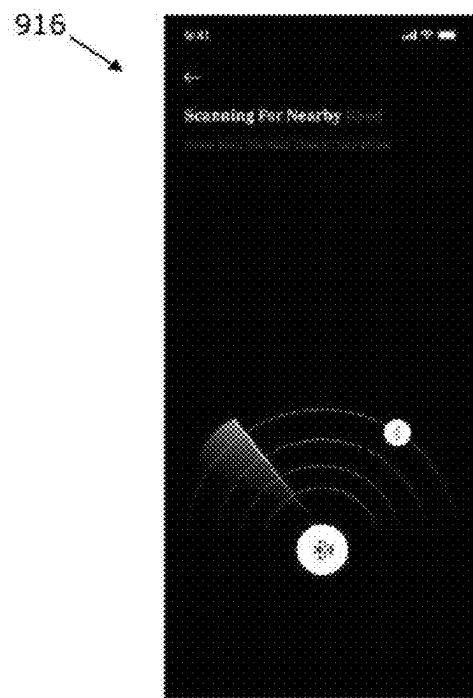


FIG. 9I

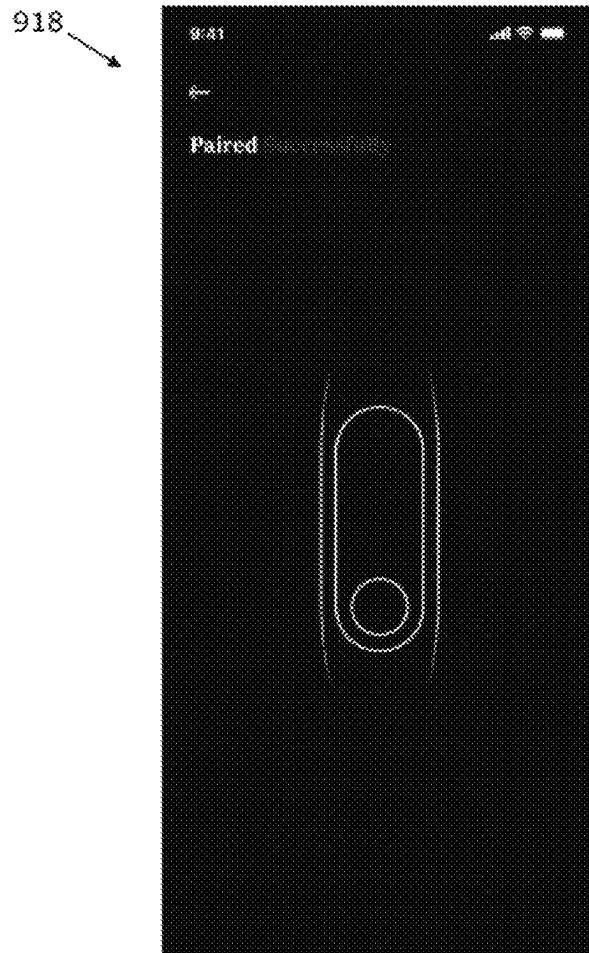


FIG. 9J

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SMART WRIST BAND**RELATED APPLICATION**

This application claims priority of provisional application 5
63/428,159 filed on Nov. 28, 2023. The entire disclosure of
this provisional application is incorporated herein by refer-
ence.

TECHNICAL FIELD

The present disclosure relates to wearable technology for
personal safety and health management, integrating emer-
gency response, health data communication, and geoloca-
tion features.

BACKGROUND

In the sphere of personal safety and health management,
the intersection of technology and individual well-being is
witnessing a paradigm shift. The pursuit of devices that can
offer instantaneous support during critical times is intensi-
fying, motivated by a collective desire for security and swift
access to medical aid. The trend is a reflection of a society
that is increasingly cognizant of the unpredictable nature of
health and safety incidents, and the necessity for immediate,
reliable response mechanisms.

Personal emergency response systems, historically, have
been static, confined to the home, and reliant on the ability
of user to reach a stationary communication device. How-
ever, as the public space has grown more complex and as
lifestyles have become more mobile, the traditional systems
are no longer sufficient. Individuals seek the assurance that
comes with having a constant, wearable companion, one that
can summon help, provide crucial health data, and ensure
their whereabouts are known, without intruding upon the
routines of daily life.

The contemporary approach to safety is proactive, that is
about having the means to signal distress with minimal
effort, the assurance that location of user can be immedi-
ately pinpointed, and that personal health details are access-
ible to responders/recipients. The connectivity allows for the seam-
less transmission of alerts and personal health profiles to
appropriate recipients, be they family, friends, or emergency
services. The system must respect user privacy while also
providing necessary information when consent is given or
when an emergency situation overrides such considerations.

The need for real-time data sharing in critical situations
has never been more apparent. As populations age and as the
desire for independent living grows, so does the need for a
system that can support the societal trends. Furthermore, the
reality of global travel and the potential for emergencies far
from home has amplified the requirement for systems that
can transcend geographical boundaries, providing peace of
mind to travelers and their loved ones.

The reliance on such systems extends beyond personal
emergencies. In the broader context of healthcare manage-
ment, the ability to track and share health metrics in real-
time can facilitate early intervention, preventing emergen-
cies before they occur. The systems, therefore, are not just
reactive but increasingly preventive.

Despite the strides made in developing user-centric tech-
nologies, the scope for enhancement is vast. The integration
of advanced sensors, predictive analytics, and adaptive
communication technologies into user assistance systems is
an ongoing journey. As the complexity of societal needs
grows, so does the imperative for reliable, and accessible

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technology solutions. Therefore, there is a need for safety
solutions, integrating emergency response, and geolocation
features.

SUMMARY

The aim of the present disclosure is to provide a user
assistance system and a method for providing user assistance
through a user assistance system comprising a wearable
device and a mobile terminal, for personal safety and health
management, integrating emergency response, health data
communication, and geolocation features.

In an embodiment, the disclosure presents a user assis-
tance system that includes a wearable device to be worn by
the user, wherein the wearable device comprises: an enclo-
sure for housing one or more electronic components,
wherein the enclosure is connected to a band to enable
wearing of the wearable device around a body part of the
user, wherein the one or more electronic components are
selected from: a display screen disposed on a top of the
enclosure, wherein the display screen displays a multidim-
ensional code; an input interface disposed on a side of the
enclosure, wherein the input interface receives a SOS input
to trigger a SOS signal; a geolocation determination unit
determines a current geolocation of the wearable device; and
a data transceiver to enable communication with a mobile
terminal; and the mobile terminal: scans the multidimen-
sional code to establish a persistence communication session
between the mobile terminal and the wearable device;
receives a user profile, wherein the user profile comprising
at least one of: at least one emergency contact details of one
or more recipients; a login credential; a medical record data;
acquires the triggered SOS signal; and triggers an emer-
gency notification to the one or more recipients, wherein the
emergency notification comprising the determined current
geolocation.

In an embodiment, the user profile is tagged with an order
history comprising one or more placed orders, wherein each
order is indexed, individually, with: a list of purchased
products or services and a vendor thereof; a loyalty data
corresponding to each of purchased product or service; and
a payment data.

In an embodiment, the loyalty information is selected
from the loyalty points, the discount coupons, or a mem-
bership status.

In an embodiment, the multidimensional code is a QR
code that encodes information related to an identity of the
user or medical information.

In an embodiment, the input interface is a button, a
touch-sensitive surface, and a voice-activated unit.

In an embodiment, the emergency notification includes an
audio data, a visual alert, or a vibration alert.

In an embodiment, the wearable device comprises a fall
detection unit that automatically triggers the SOS signal.

In an embodiment, the SOS signal includes a pre-recorded
audio message that is transmitted to the one or more recipi-
ents.

In an embodiment, the wearable device further includes a
camera capable to capture multi-media data.

In an embodiment, a server arrangement analyzes the
order history and flags a suspicious transaction.

In an embodiment, the server arrangement uses a machine
learning algorithm to personalize one or more shopping
suggestions based on the order history.

In an embodiment, the disclosure pertains to a method for
assisting a user, the method comprising: arranging of one or
more electronic components in an enclosure of a wearable

device, wherein the enclosure is connected to a band to enable wearing of the wearable device around a body part of a user, wherein the one or more electronic components enable: displaying a multidimensional code on a display screen, wherein the display screen is disposed on a top of the enclosure; receiving a SOS input on a user interface, to trigger a SOS signal; determining a current geolocation of the wearable device; communication with a mobile terminal, through a data transceiver; scanning the multidimensional code, through the mobile terminal, to establish a persistence communication session between the mobile terminal and the wearable device; receiving a user profile, at the mobile terminal; acquiring the triggered SOS signal, at the mobile terminal; and triggering, through the mobile terminal, an emergency notification to the one or more recipients, wherein the emergency notification comprising the determined current geolocation.

In an embodiment, the wearable device generates and emits a unique sound to assist search and rescue teams in locating the user.

In an embodiment, the wearable device generates an augmented reality display to provide a guidance instruction to the user during emergencies.

In an embodiment, the multidimensional code is a QR code that encodes information related to an identity of the user or medical information.

In an embodiment, the input interface is a button, a touch-sensitive surface, and a voice-activated unit.

BRIEF DESCRIPTION OF THE DRAWINGS

The summary above, as well as the following detailed description of illustrative embodiments, is better understood when read in conjunction with the appended drawings. For the purpose of illustrating the present disclosure, exemplary constructions of the disclosure are shown in the drawings. However, the present disclosure is not limited to specific methods and instrumentalities disclosed herein.

Embodiments of the present disclosure will now be described, by way of example only, with reference to the following diagrams.

FIG. 1 illustrates a user assistance system, in accordance with the embodiments of the present disclosure;

FIG. 2 illustrates a method for providing user assistance through a user assistance system comprising a wearable device and a mobile terminal, in accordance with the embodiments of the present disclosure;

FIG. 3 illustrates a block diagram of a system for providing assistance to a user, in accordance with another embodiment of the present disclosure;

FIG. 4 illustrates a perspective view of a wearable monitoring device, in accordance with an embodiment of the present disclosure;

FIG. 5 depicts a perspective view of a smart band, in accordance with an embodiment of the present disclosure;

FIG. 6 depicts an exploded view of the smart band, in accordance with an embodiment of the present disclosure;

FIG. 7 (FIG. 7A to FIG. 7B) shows a perspective view of an enclosure, in accordance with an embodiment of the present disclosure;

FIG. 8 illustrates a block diagram illustrating electrical communication within a wristband, in accordance with an embodiment of the present disclosure; and

FIG. 9A-9J illustrates the various graphical user interfaces (GUIs) associated with an application for providing assistance to a user, in accordance with an embodiment of the present disclosure.

DETAILED DESCRIPTION OF EMBODIMENTS

Various objects, features, aspects and advantages of the inventive subject matter will become more apparent from the following detailed description of preferred embodiments, along with the accompanying drawing figures in which like numerals represent like components.

The following discussion provides many example embodiments of the inventive subject matter. Although each embodiment represents a single combination of inventive elements, the inventive subject matter is considered to include all possible combinations of the disclosed elements. Thus, if one embodiment comprises elements A, B, and C, and a second embodiment comprises elements B and D, then the inventive subject matter is also considered to include other remaining combinations of A, B, C, or D, even if not explicitly disclosed.

In some embodiments, the numbers expressing quantities of ingredients, properties such as concentration, reaction conditions, and so forth, used to describe and claim certain embodiments of the invention are to be understood as being modified in some instances by the term “about.” Accordingly, in some embodiments, the numerical parameters set forth in the written description and attached claims are approximations that can vary depending upon the desired properties sought to be obtained by a particular embodiment. In some embodiments, the numerical parameters should be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. Notwithstanding that the numerical ranges and parameters setting forth the broad scope of some embodiments of the invention are approximations, the numerical values set forth in the specific examples are reported as precisely as practicable. The numerical values presented in some embodiments of the invention may contain certain errors necessarily resulting from the standard deviation found in their respective testing measurements.

As used in the description herein and throughout the claims that follow, the meaning of “a,” “an,” and “the” includes plural reference unless the context clearly dictates otherwise. Also, as used in the description herein, the meaning of “in” includes “in” and “on” unless the context clearly dictates otherwise.

The recitation of ranges of values herein is merely intended to serve as a shorthand method of referring individually to each separate value falling within the range. Unless otherwise indicated herein, each individual value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided with respect to certain embodiments herein is intended merely to better illuminate the invention and does not pose a limitation on the scope of the invention otherwise claimed. No language in the specification should be construed as indicating any non-claimed element essential to the practice of the invention.

Groupings of alternative elements or embodiments of the invention disclosed herein are not to be construed as limitations. Each group member can be referred to and claimed individually or in any combination with other members of the group or other elements found herein. One or more members of a group can be included in, or deleted from, a group for reasons of convenience and/or patentability. When any such inclusion or deletion occurs, the specification is

herein deemed to contain the group as modified thus fulfilling the written description of all Markush groups used in the appended claims.

The following detailed description illustrates embodiments of the present disclosure and ways in which they can be implemented. Although some modes of carrying out the present disclosure have been disclosed, those skilled in the art would recognize that other embodiments for carrying out or practicing the present disclosure are also possible.

FIG. 1 illustrates a user assistance system **100** (interchangeably referred as system **100**), in accordance with the embodiments of the present disclosure. The system **100** comprises a wearable device **102**, a mobile terminal **110** and other known components. In an embodiment, the user assistance system **100** comprises the wearable device **102** that is worn by the user on a body part, facilitating constant proximity to the user for the purpose of providing assistance. The wearable device **102** comprises an enclosure **104**, which can house one or more electronic components **106-A** (individually referred as **106-A1**, **106-A2**, **106-A3**, . . . **106-AN**). The one or more electronic components **106-A** provides computational power, sensory data processing, and connectivity features essential for the operation of the user assistance system **100**.

In an embodiment, the enclosure **104** is fabricated from materials selected for durability, comfort, and a lightweight profile to ensure ease of use over extended periods. The enclosure **104** provides protection to the electronic components **106-A** contained within, safeguarding against physical damage and environmental factors that could disrupt the functionality of the wearable device **102**. In an embodiment, the enclosure **104** is connected to a band **108** that secures the wearable device **102** around the body part of the user. The band **108** is adjustable to accommodate various body sizes and shapes, ensuring that the wearable device **102** remains securely positioned during use. The band **108** is constructed from materials that offer a balance of flexibility and strength, allowing for a comfortable fit while preventing slippage or displacement of the wearable device **102**.

In an embodiment, the wearable device **102** is made to be unobtrusive and ergonomically designed, allowing users to wear the wearable device **102** without hindrance to their daily activities. The contours and dimensions of the enclosure **104** are optimized to conform to the anatomy of the body part (e.g., wrist, finger, arm, ear, forearm, etc.) on which the wearable device **102** is to be worn, thereby enhancing the user experience through improved wearability.

In an embodiment, the electronic components **106-A** housed within the enclosure **104** include, but are not limited to, a processor, memory, communication modules, sensors, and a power source. The processor can execute instructions from a set of executable routines stored within the memory, processing sensory data received from the sensors, and managing communications facilitated by the communication modules.

In an embodiment, the power source contained within the enclosure **104** supplies power to the electronic components **106-A**. The power source (e.g., Lithium-Ion battery) can allow for recharging, incorporating safety mechanisms to protect against overcharging and power surges.

In an embodiment, the enclosure **104** comprises a display screen **106-A1** that is one of the one or more electronic components **106-A** incorporated into the wearable device **102**. The display screen **106-A1** presents information in a visually accessible manner, wherein the display screen **106-A1** exhibits a multidimensional code (e.g., QR code, bar-

code etc. shown as **408** in FIG. 4). The multidimensional code displayed by the display screen **106-A1** conveys data in a graphical format which can be easily scanned and interpreted by compatible reading devices. The multidimensional code can be a means through which the user can interface with various aspects of the user assistance system **100**, facilitating seamless interactions within the ecosystem of the services of system **100**. The placement and functional integration of the display screen **106-A1** within the wearable device **102** exemplify the approach to ergonomic and efficient design that the user assistance system **100** endeavors to provide, ensuring that information is readily available to the user with minimal effort and maximum convenience.

In an embodiment, the enclosure **104** comprises an input interface **106-A2** that is positioned on a side of the enclosure **104**. The input interface **106-A2** receives an input signal indicative of an emergency condition, commonly referred to as an SOS input. Upon actuation by the user, the input interface **106-A2** initiates a sequence of operations collectively recognized as an SOS signal that facilitates rapid response actions, which may include but are not limited to, notifying emergency services, alerting pre-stored contacts, or activating location services to determine the geographic position of the user in distress. The inclusion of the input interface **106-A2** on the wearable device **102** ensures that assistance is readily accessible through a simple, user-activated mechanism, thereby enhancing the safety and well-being of the user in situations where immediate aid is required.

In an embodiment, the wearable device **102** comprises a geolocation determination unit **106-A3** to determine geolocation of the wearable device **102** by utilizing global positioning technology. The geolocation determination unit **106-A3** ensures that the wearable device **102** offers location-based services. The services include, but are not limited to, real-time user tracking for enhanced safety, historical visited location tracing, efficient navigational assistance, and the provision of geographically pertinent information. The geolocation determination unit **106-A3** delivers continuous locational updates aids to the functionality of the user assistance system **100**. The geolocation determination unit **106-A3** enables the user assistance system **100** to deliver tailored support and information of the user, thereby enhancing the overall effectiveness and user experience of the system **100**.

In an embodiment, the one or more electronic components **106-A** includes a data transceiver **106-A4** that is selected to facilitate communication with the mobile terminal **110**. The data transceiver **106-A4** is configured to transmit and receive data packets, thereby enabling a seamless exchange of information between the user assistance system **100** and the mobile terminal **110**. The user assistance system **100** enables communication with the mobile terminal **110** for various functionalities that include, but are not limited to, transmitting notifications to the user, receiving user inputs, and exchanging operational data. The data transceiver **106-A4** ensures that the user assistance system **100** maintains a connectivity framework capable of supporting a multitude of communication protocols, which may include cellular, Wi-Fi, Bluetooth, NFC, or any combination thereof. The data transceiver **106-A4** upholds data integrity and security standards, ensuring that all communications between the user assistance system **100** and the mobile terminal are conducted in a secure and reliable manner, safeguarding user data against unauthorized access and breaches.

In an embodiment, the mobile terminal **110** performs a scanning operation on the multidimensional code. Upon

successful scanning of the multidimensional code, the mobile terminal **110** is further configured to establish a persistent communication session with the wearable device **102**. The persistent communication session is initiated and maintained by the mobile terminal **110** to facilitate continuous and uninterrupted data exchange between the mobile terminal **110** and the wearable device **102**. The persistence of the communication session ensures that, once established, the session remains active, allowing for real-time interaction and data synchronization between the mobile terminal **110** and the wearable device **102**, without the need for re-establishment of the communication link. The communication session thus provides a seamless user experience, whereby the functionality and services offered by the wearable device **102** can be accessed, controlled, or monitored via said mobile terminal **110**.

In an embodiment, the mobile terminal **110** receives a user profile, which may include, variety of information such as personal information pertinent to the security and health of user. Among the data received by the mobile terminal **110**, at least one emergency contact detail (e.g., email address, mobile number, social media account details etc.) is included for one or more designated recipients (e.g., caregiver, doctor, nurse, family member, community center, etc.) ensuring that in the event of an emergency, communication can be established to facilitate immediate assistance. Additionally, the user profile is structured to encompass login credentials, which are essential for securing user access and maintaining the integrity of the system **100** and the privacy of the data contained therein. Furthermore, the user profile includes medical record data (e.g., co-morbid disease condition, allergic information, current medication intake etc.), which is paramount in instances where medical intervention is required, allowing healthcare professionals rapid access to vital health information that can inform treatment decisions. The collected data within the user profile is integral to the functionality of the user assistance system **100**, enabling the mobile terminal **110** to act as a touchpoint in the provision of user assistance.

In an embodiment, upon activation by user, the SOS signal, being a predetermined communication initiated during instances of emergency or distress, is captured by the mobile terminal **110**. The mobile terminal **110**, upon acquisition of the SOS signal, engages in a series of predefined operations aimed at facilitating rapid and effective assistance. The predefined operations may include, but are not limited to, transmitting the SOS signal to designated emergency contacts, emergency services, and/or other entities within a response network. Additionally, the mobile terminal **110** collates and transmits ancillary data pertinent to the emergency situation, which may encompass the geographical location of the mobile terminal **110**, the timestamp at which the SOS signal was triggered, as well as any other contextual information that may be preconfigured by the user or inferred by the system **100**. The transmission of aforesaid data is executed via established communication networks, ensuring that the SOS signal, along with the accompanying data, is relayed promptly and securely. Furthermore, the mobile terminal **110** may initiate additional protocols concurrently with the transmission of the SOS signal, including the activation of audio and visual signals to attract attention or to deter threats in the vicinity of the user. The integrated functionalities of such mobile terminal **110** coalesce to form a response framework within the user assistance system **100**, providing an efficient mechanism for users to signal distress and obtain assistance with heightened efficacy and reliability.

In an embodiment, upon activation, the mobile terminal **110** dispatches an emergency notification to one or more predetermined recipients. The content of the emergency notification includes, but may not be limited to, the current geolocation, which is ascertained by the wearable device **102** and mobile terminal **110** at the time of the emergent event. The geolocation data is determined through the utilization of geolocation determination unit **106-A3**, ensuring that the coordinates are captured and relayed. Such features are vital for providing immediate and accurate locational information during crisis situations, thereby facilitating prompt response and assistance. The process by which the emergency notification is generated and disseminated is automated, requiring minimal user interaction, which is of paramount importance in situations where the user may be incapacitated or otherwise hindered from manually transmitting alerts.

In an embodiment, the user assistance system **100** may include the user profile that is tagged with an order history that comprises one or more placed orders. Each of the orders is indexed individually with a list of purchased products or services along with the respective vendor of each product or service. Further included in each order is loyalty data corresponding to each of the purchased products or services. Additionally, payment data associated with each of the orders is recorded. The loyalty data may include points accrued, membership status, or other vendor-specific loyalty metrics that are pertinent to the engagement of user with the vendor. The payment data may encompass the transaction amounts, chosen payment methods, transaction dates, and other relevant financial details that facilitate a view of the purchasing activities of the user. The amalgamation of indexed order history, loyalty data, and payment data in the user profile enables a tailored user experience, allowing for enhanced customer service and personalized marketing opportunities by the user assistance system **100**.

In an embodiment, the loyalty information may be characterized by comprising at least one selected from the group consisting of: loyalty points indicative of a value accumulated based on previous transactions or interactions; discount coupons which are applicable to future transactions and may be redeemed for goods, services, or advantageous financial terms; and membership status reflecting the current standing of user within a loyalty program, said status affording the user preferential access to services, promotions, or information. The loyalty information is stored, managed, and updated within the system **100**, wherein upon authentication of the user, the loyalty information is retrieved and utilized to customize the user experience, offering personalized incentives and enhancing user engagement with the system **100**. The system **100** interacts with the mobile terminal **110** to ensure real-time accuracy and relevance of the loyalty information presented to the user, thereby fostering a seamless and customized interaction with the system **100** based on the loyalty-related data of user.

In an embodiment, the multidimensional code may be implemented as a quick response (QR) code, which encodes information pertinent to the user utilizing the system **100**. The QR code may incorporate name, contact information, and any identifying numbers or credentials that may be utilized to uniquely distinguish the user. In addition, the QR encodes medical information of user. Such medical information can range from the medical history of user, including past diagnoses and treatments, to current medical conditions that may require immediate attention or consideration, such as allergies, ongoing medications, or specific medical directives that are essential in emergency situations. The encoded

information within the QR code can be readily accessed and deciphered by the mobile terminal **110**, thereby facilitating rapid identification and the provision of personalized assistance. The aforementioned feature of the user assistance system **100** ensures that in instances where the user is unable to communicate effectively, or identification is otherwise impeded, the essential information can still be obtained through the scanning of the QR code by the mobile terminal **110**. The QR code thus serves as a secure and efficient means of storing and transmitting sensitive user information, which can be imperative in providing timely and accurate support to the user by first responders, medical personnel, or any other entities requiring access to the personal or medical data of user.

In an embodiment, the input interface **106-A2** may be selected from a button, a touch-sensitive surface, and a voice-activated unit (comprising microphone and speech-to-text conversion unit). The button may be integrated into the user assistance system **100** to enable tactile engagement by users, providing a simple and effective means for initiating interaction with the system **100**. Additionally, the touch-sensitive surface may be incorporated to capture gestures and touches, facilitating a responsive user input method that accommodates a range of commands and controls. The touch-sensitive surface can recognize varying degrees of pressure and patterns of touch, thereby allowing for a diverse array of input possibilities. Complementing the tactile components of the input interface **106-A2**, the voice-activated unit interprets auditory signals from users. The voice-activated unit employs advanced speech recognition algorithms that are capable of processing natural language commands and converting them into operational directives for such user assistance system **100**. The voice-activated unit detects spoken words, phrases, and queries, enabling hands-free operation, and offering an accessible mode of interaction for users. The incorporation of the voice-activated unit is beneficial in scenarios where manual interaction is impeded or impractical.

In an embodiment, the emergency notification may employ a multimodal alert system, providing notifications through audio data, visual representation, and vibration alert, each serving to ensure that the recipient receives a prompt and clear indication of an emergency. The audio alert is configured to emit a distinct sound at a volume calculated to capture the attention of user even in a noisy environment. Concurrently, the visual alert is manifested in the form of flashing lights or on-screen messages displayed on the mobile terminal **110** to be discernible even at a distance or in bright conditions. Additionally, the vibration alert provides a tactile warning through rhythmic or pulsating vibrations, which can be particularly effective in scenarios where the user may not be able to rely on auditory or visual alerts, such as in loud surroundings or in situations where visual attention is focused elsewhere. The combination of the alert modalities ensures an efficient mechanism for conveying emergency notifications, thereby enhancing the safety and responsiveness of the user.

In an embodiment, the wearable device **102** may comprise a fall detection unit that is arranged to autonomously monitor the stability and orientation of user. Upon detection of a fall, characterized by a sudden change in altitude and orientation indicative of a user becoming prone, the fall detection unit initiates an SOS signal. The activation of the SOS signal is immediate, ensuring prompt communication of the emergency. Once activated, the SOS signal is transmitted to the mobile terminal **110**, which is operatively coupled to the wearable device **102**. The mobile terminal

110, upon receipt of the SOS signal, proceeds to execute a series of predefined actions, which may include, but are not limited to, alerting emergency services, notifying predefined contacts within the personal network of user, and providing location data that is crucial for expedited assistance. The execution of the actions by the mobile terminal **110** is conducted in a manner that prioritizes urgency and accuracy, thereby enhancing the safety and security of the user in situations where the user may be incapacitated and unable to manually seek assistance.

In an embodiment, the SOS signal may be generated by the mobile terminal **110**, which includes a pre-recorded audio message stored within the memory of the mobile terminal **110**. Upon activation of the SOS signal by the user, the pre-recorded audio message is transmitted to a plurality of pre-designated emergency contacts/recipients. The transmission of the pre-recorded audio message is facilitated by a data transceiver **106-A4** of the mobile terminal **110**, wherein the data transceiver **106-A4** communication module is interfaced with various communication networks to ensure transmission of the SOS signal. The transmission ensures that the emergency contacts/recipients are alerted to the distress situation of user. The system **100** further ensures that the transmission of the SOS signal is executed with the highest priority over other communications to expedite the delivery of the pre-recorded audio message to the intended recipients, thus enhancing the responsiveness and effectiveness of the user assistance system **100**.

In an embodiment, the wearable device **102** may comprise a camera for capturing multi-media data (e.g., photographic and video content). The captured content can be transmitted to the mobile terminal **110**. The transmission of the visual content from the wearable device **102** to the mobile terminal **110** enables the user to share visual data, thereby enhancing utility of the user assistance system **100**. The visual content, once transmitted to the mobile terminal **110**, can be further processed, analyzed, or stored, depending on user requirements.

In an exemplary embodiment, the system **100** can comprise a server arrangement (depicted as **304** in FIG. 3) that can enhance security and reliability of financial transactions by analyzing order history and identifying suspicious transactions. The server arrangement can include data analysis tools (e.g., machine learning and artificial intelligence) to recognize patterns typical of legitimate transactions and those indicative of fraudulent activities. The server continuously receives and analyzes data (e.g., transaction amount, frequency, geographical location, and other relevant metadata) related to new transactions, comparing them against the order history of the account in question. If a transaction significantly deviates from established patterns in the order history, the server can flag particular transaction as suspicious and trigger notification to user/financial institution/vendor/payment gateway etc.

The system **100** can also be configured to take automatic actions, like freezing the account or transaction, initiating refund request, and the like. The system **100** can enable enhancement of transaction security, by analyzing order history to identify anomalies, thereby reducing the risk of fraudulent transactions.

In an aspect, the server arrangement can utilize machine learning algorithms to provide personalized shopping suggestions based on order history to improve shopping experience. The machine learning algorithms can identify preferences and habits by examining past purchases, including product types, brands, purchase frequency, and spending patterns. Based on the analysis, the machine learning algo-

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rhythm generates shopping suggestions tailored to the user's preferences and habits. The personalized shopping suggestions enhance user experience and boosts vendor revenue.

In an embodiment, the wearable device **102** can be equipped with a sound-generating unit (e.g., speaker) to generate sound that can be easily recognizable and distinguishable from ambient noises. The sound is customized to carry over long distances and through various terrains, ensuring that sound can be heard by search and rescue team. The sound generation can be activated manually by the user in an emergency or automatically triggered under certain conditions, such as when the user is immobile for an extended period or when the device detects a fall. The sound pattern can be customized based on the user's profile or the nature of the emergency, providing specific information to the rescuers about the situation.

In another aspect, the wearable device **102** includes an AR display mechanism, which can be a built-in screen or a projection unit that interfaces with smart glasses or similar technology. The AR display provides visual guidance instructions, for an instance, e.g., navigation aid to the nearest exit in a building, the location of emergency resources, or instructions for self-help in medical emergencies. The AR display offers a more intuitive and effective way for users to understand and follow emergency instructions.

FIG. 2, illustrates the flow diagram of method **200** for assisting a user using a wearable device, in accordance with embodiment of present disclosure. The step **210** involves placing various electronic components within enclosure **104** of the wearable device **102**, which can be worn around a body part, like a wrist, facilitated by a band. The step **204** involves displaying a multidimensional code (e.g., QR code) on a display screen of the wearable device **102**. The screen is disposed on a top of the enclosure **104**. The multidimensional code can be used for various purposes, including pairing of wearable device **102** or information sharing.

The step **206** comprises receiving, through user interface (e.g., button, touch sensitive screen etc.) an SOS signal input from user. The user can trigger SOS signal for emergency situations to provide a quick distress signal. The step **208** involves determination of geolocation of wearable device **102** GPS or similar technology. The determined geolocation would be vital for emergency situations.

At step **208**, user can use a mobile terminal **110** (e.g., smartphone) to scan the displayed multidimensional code on the wearable device **102**, for establishing a persistent communication session. Scanning of multidimensional code can be used for a variety of purposes, including initial setup or emergency services. At step **210**, wearable device **102** communicates with mobile terminal **110** via a data transceiver **106-A4** to enable exchange of data (e.g., determined geo location) and control commands (e.g., SOS command).

The step **212** involves receiving a user profile, at the mobile terminal **110**. The user profile includes essential information such as emergency contact details of one or more recipients, login credentials, and relevant medical record data.

The step **216**, the mobile terminal **110** captures and processes the triggered SOS signal from the wearable device **102**, recognizing the urgency of the situation. At step **218**, utilizing the acquired information, the mobile terminal **110** promptly triggers an emergency notification. The emergency notification includes details such as the determined current geolocation of the wearable device **102**. The emergency notification is then transmitted to the designated recipients, ensuring swift response and assistance.

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FIG. 3 illustrates a block diagram of a system **300** for providing an assistance to a user, in accordance with another embodiment of the present disclosure. The system **300**, incorporates the wearable device **102** and the mobile terminal **110** (same as FIG. 1) and operatively couples with a third-party server arrangement **304**, through internet-based long-range wireless communication. The server arrangement **304** is associated with vendors offering various products and services. The server arrangement **304** conveys promotional material to the wearable device **102**, thus personalizing the user experience with relevant sales and discount information.

In an embodiment, the third-party server arrangement **304** manages loyalty information derived from the past transactions of user. By processing purchase-related data received from the wearable device **102**, the third-party server arrangement **304** computes loyalty points that are then relayed back to the user, reflecting their accrued benefits. The loyalty information encompasses data such as total loyalty points and current membership status, which is subsequently made available upon request.

In embodiment, a database arrangement **306** serves as a repository for a user profile that includes personal, medical, and emergency recipient/contact information. Database arrangement **306** is continually updated with any changes, such as alterations to security PINs, made through the mobile terminal **110**. The database arrangement **306** also tracks loyalty information related to the dealings of user with multiple vendors, updating the loyalty point tally after each transaction.

In an embodiment, the system **300** comprises a server arrangement **308**, which is in communication with the wearable device **102** and the mobile terminal **110**. The server arrangement **308** processes the current location of user that is important for identifying the locale vendor and current shopping information, such as items added to a digital shopping cart. The server arrangement **308** calculates updated billing costs by considering the loyalty information, leading to discounted prices at the point of sale, which are then communicated back to the mobile terminal **110** and displayed on the wearable device **102** as a QR code for easy scanning and payment.

FIG. 4 illustrates a perspective view of a wearable monitoring device **400** (that is the wearable device **102** of FIG. 1), in accordance with an embodiment of the present disclosure. The wearable monitoring device **400**, as depicted, integrates a wrist band **402** (that is band **108** in FIG. 1), which fits to the user's body part comfortably, thereby allowing for ease of wearability of the wearable monitoring device **400**. In addition to the band **402**, the wearable monitoring device **400** comprises an light emitting diode (LED) display **404** (identifiable as display screen **106-A1** from FIG. 1), which serves the dual purpose of showing the current time **406** and a unique code **408**, the latter of which can be used for various authentication or identification purposes. The LED display **404** thus acts as a convenient interface for the wearer, presenting essential information briefly. Complementing the features is an emergency button **410** (that is the input interface **106-A2** shown in FIG. 1), which provides users with a quick and efficient way to signal distress. The emergency button **410** is indicative of the utility of wearable monitoring device **400** in critical situations, allowing the wearer to activate an emergency protocol (such as SOS signal) with a single press. Such a feature is particularly invaluable in circumstances where immediate assistance is required, ensuring that users have a readily accessible means to alert others when in need.

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FIG. 5 depicts a perspective view of a smart band 500 (such as the wearable monitoring device 400 of FIG. 4 and the wearable device 102 of FIG. 1), in accordance with an embodiment of the present disclosure. As depicted, the smart band 500 comprises a detachable band 502, featuring a mechanism that allows smart band 500 to be opened, thereby facilitating the ease of removing the smart band 500, as required. This ensures that the smart band 500 can be securely fastened around the wrist or other body parts, and equally conveniently removed, catering to the need of user for flexibility and comfort. The detachable nature of band 502 enhances the user experience by allowing for quick and effortless adjustments to fit and enables the possibility of customizing or replacing the band 502 entirely, without needing to replace the entire detachable band 502. Such features are particularly advantageous for users who may wish to interchange bands of different colors or materials to match their personal style or for specific activities, thereby extending the versatility and functional appeal of the smart band 500.

FIG. 6 depicts an exploded view of the smart band 500 shown in FIG. 5, in accordance with an embodiment of the present disclosure. The smart band 500, as depicted, features an enclosure 504 (that is the enclosure 104 presented in FIG. 1) that is arranged to house the electronic components 106-A of smart band 500. Atop the enclosure 504, liquid crystal display (LCD) display 506, (such as the LCD display 404 and display screen 106-A1), serves as the visual interface for the user. The LCD display 506 is positioned for optimal visibility and ease of interaction. Complementing the visual component, the smart band 500 comprises an emergency button 508 (that is emergency button 410 and input interface 106-A2), which is integrated into the right side of the enclosure 504. The placement of the emergency button 508 ensures the accessibility in urgent situations, enabling the wearer to initiate an emergency protocol (that is SOS signal) with a simple press. The smart band 500 provides visual output through the LCD display 506 and user input via the emergency button 508, to provide a seamless and intuitive user experience.

FIG. 7 (FIG. 7A to FIG. 7B) shows a perspective view of an enclosure 700 (such as the enclosure 504 of FIG. 6 and the enclosure 104 presented in FIG. 1), in accordance with an embodiment of the present disclosure. The enclosure 700, as depicted, includes an outer casing 702 constructed from materials such as plastic, which serves to house the electronic components 106-A of a wristband (such as the smart band 500 illustrated in FIGS. 5-6), effectively ensuring both water and airtight protection. Integral to the enclosure 700 are components 106-A such as an LCD display 704 (such as LCD display 506, the LCD display 404 and display screen 106-A1) and an emergency button 706 (such as emergency button 508, emergency button 410 and input interface 106-A2), which facilitates user interaction and display functionalities. In addition, the enclosure 700 comprises an LED 708 that, in conjunction with a light guide 710, creates an illuminated band 712 situated below the LCD display 704, as well as an illuminated band 714 positioned above the LCD display 704, enhancing the visibility and aesthetics of the wristband. The enclosure 700 also integrates a printed circuit board 716, which is operationally connected to the emergency button 706. Such connection is vital in activating the functionality of the emergency button 706, allowing users to initiate an emergency protocol, when necessary.

FIG. 8 illustrates a block diagram 800 illustrating electrical communication within a wristband (such as the smart band 500 of FIG. 5, the wearable monitoring device 400 of

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FIG. 4 and the wearable device 102 of FIG. 1), in accordance with an embodiment of the present disclosure. The configuration indicates that a 5V DC input electrical supply is directed to a battery charger (such as, through wall-mounted electrical supply), which is responsible for charging a lithium polymer (LiPo) battery 718. Subsequently, the electrical charge accumulated within the LiPo battery 718 is distributed via the power source to various components 106-A including an emergency button 706 (such as emergency button 508, emergency button 410, and input interface 106-A2), an LCD 704 (such as, LCD display), and a data transceiver 106-A4. The configuration ensures that the emergency button facilitates quick user response in exigent situations, while the LCD display 704 maintains the function to exhibit vital user information. Additionally, the data transceiver 106-A4 handles wireless communication requirements.

FIG. 9A-9J illustrates the various graphical user interfaces (GUIs) 900-918 associated with an application for providing an assistance to a user, in accordance with an embodiment of the present disclosure. The disclosed application (interchangeably referred as app), exemplified on a mobile device such as the mobile terminal 110 depicted in FIGS. 1, offers a graphical user interface (GUI) 900 as illustrated in FIG. 9A, which enables users to register for the app. Upon completion of registration, the users log-in into their accounts via GUI 902, as delineated in FIG. 9B. The application facilitates the entry of user details through GUIs 904 and 906, as seen in FIGS. 9C and 9D, respectively. The information entry culminates the creation of user profiles, which are displayed as profile 908 and 910 in FIGS. 9E and 9F. Following the establishment of their profiles, users are presented with the option to subscribe to a membership plan through GUI 912, showcased in FIG. 9G. Post subscription, users can pair their smart wristband with the mobile terminal utilizing a BLE connection and the interactive steps provided by GUIs 914-918 as seen in FIGS. 9H-9J. Such sequence of interfaces guides the user from initial engagement with the application to the full integration of their wearable device 102, enhancing the user experience and the utility of the system.

In an aspect, present disclosure provides a computer-readable storage medium is equipped with data and instructions, which, upon execution, perform method 200 designed to assist a user. The method 200 encompasses a series of steps executed by a processor coupled to a non-transitory storage device. The method 200 involves the utilization of multiple electronic components 106-A1 situated in the enclosure 104 of the wearable device 102. This enclosure is connected to a band, enabling the device to be worn around a user's body part. The electronic components 106-A1 are configured to enable several functionalities.

The foregoing description of the specific embodiments will so fully reveal the general nature of the embodiments herein that others can, by applying current knowledge, readily modify and/or adapt for various applications such specific embodiments without departing from the generic concept, and, therefore, such adaptations and modifications should and are intended to be comprehended within the meaning and range of equivalents of the disclosed embodiments. It is to be understood that the phraseology or terminology employed herein is for the purpose of description and not of limitation. Therefore, while the embodiments herein have been described in terms of preferred embodiments, those skilled in the art will recognize that the embodiments herein can be practiced with modification within the spirit and scope of the embodiments as described herein.

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It should be apparent to those skilled in the art that many more modifications besides those already described are possible without departing from the inventive concepts herein. The inventive subject matter, therefore, is not to be restricted except in the spirit of the appended claims. Moreover, in interpreting both the specification and the claims, all terms should be interpreted in the broadest possible manner consistent with the context. In particular, the terms “comprises” and “comprising” should be interpreted as referring to elements, components, or steps in a non-exclusive manner, indicating that the referenced elements, components, or steps may be present, or utilized, or combined with other elements, components, or steps that are not expressly referenced.

What is claimed is:

1. A user assistance system, the system comprising:
a wearable device to be worn by the user, wherein the wearable device comprises:
an enclosure for housing one or more electronic components, wherein the enclosure is connected to a band to enable wearing of the wearable device around a body part of the user, wherein the one or more electronic components are selected from:
a display screen disposed on a top of the enclosure, wherein the display screen displays a multidimensional code;
an input interface disposed on a side of the enclosure, wherein the input interface receives a SOS input to trigger a SOS signal;
a geolocation determination unit determines a current geolocation of the wearable device; and
a data transceiver to enable communication with a mobile terminal; and
the mobile terminal:
scans the multidimensional code to establish a persistence communication session between the mobile terminal and the wearable device;
receives a user profile, wherein the user profile comprising at least one of:
at least one emergency contact details of one or more recipients;
a login credential;
a medical record data;
acquires the triggered SOS signal; and
triggers an emergency notification to the one or more recipients, wherein the emergency notification comprising the determined current geolocation.
2. The user assistance system of claim 1, wherein the user profile is tagged with an order history comprising one or more placed orders, wherein each order is indexed, individually, with:
a list of purchased products or services and a vendor thereof;
a loyalty data corresponding to each of purchased product or service; and
a payment data.
3. The user assistance system of claim 1, wherein the loyalty information is selected from the loyalty points, the discount coupons, or a membership status.
4. The user assistance system of claim 1, wherein the multidimensional code is a QR code that encodes information related to an identity of the user or medical information.

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5. The user assistance system of claim 1, wherein the input interface is a button, a touch-sensitive surface, and a voice-activated unit.

6. The user assistance system of claim 1, wherein the emergency notification includes an audio data, a visual alert, or a vibration alert.

7. The user assistance system of claim 1, wherein the wearable device comprises a fall detection unit that automatically triggers the SOS signal.

8. The user assistance system of claim 1, wherein the SOS signal includes a pre-recorded audio message that is transmitted to the one or more recipients.

9. The user assistance system of claim 1, wherein the wearable device further includes a camera capable to capture multi-media data.

10. The user assistance system of claim 1, comprising a server arrangement to analyze the order history and flags a suspicious transaction.

11. The user assistance system of claim 1, wherein the server arrangement uses a machine learning algorithm to personalize one or more shopping suggestions based on the order history.

12. A method for assisting a user, the method comprising:
arranging of one or more electronic components in an enclosure of a wearable device, wherein the enclosure is connected to a band to enable wearing of the wearable device around a body part of a user, wherein the one or more electronic components enable:
displaying a multidimensional code on a display screen, wherein the display screen is disposed on a top of the enclosure;
receiving a SOS input on a user interface, to trigger a SOS signal;
determining a current geolocation of the wearable device;
scanning the multidimensional code, through the mobile terminal, to establish a persistence communication session between the mobile terminal and the wearable device;
communication with a mobile terminal, through a data transceiver;
receiving a user profile, at the mobile terminal;
acquiring the triggered SOS signal, at the mobile terminal; and
triggering, through the mobile terminal, an emergency notification to the one or more recipients, wherein the emergency notification comprising the determined current geolocation.

13. The method of claim 12, wherein the wearable device generates and emits a unique sound to assist search and rescue teams in locating the user.

14. The method of claim 12, wherein the wearable device generates an augmented reality display to provide a guidance instruction to the user during emergencies.

15. The method of claim 12, wherein the multidimensional code is a QR code that encodes information related to an identity of the user or medical information.

16. The method of claim 12, wherein the input interface is a button, a touch-sensitive surface, and a voice-activated unit.

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